THERMOCATOR DIGITAL TWIN TEST REPORT

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PURPOSE

This report summarizes the local build, automated test, and runtime communication checks performed for the Thermocator ROS 2 digital twin workspace. The goal was to verify that the project builds successfully, that its automated test suite passes, and that the digital twin running on ‘ROS_DOMAIN_ID=1‘ can send decision outputs into the robot-side stack on ‘ROS_DOMAIN_ID=38‘.

ENVIRONMENT

* Workspace: ‘/home/robocup/cbl-digital-twins/jazzy_tb3_packages‘
* Validation environment: Docker
* Docker image: ‘turtlebot3_ws:latest‘
* ROS domains:
* ‘Domain 38‘: robot-side stack / local simulation
* ‘Domain 1‘: digital twin Gazebo + independent Thermocator stack

The host machine did not have a native ROS 2 Jazzy installation at ‘/opt/ros/jazzy/setup.bash‘, so validation was performed in the provided Docker environment.

BUILD AND AUTOMATED TEST RESULTS

The workspace was built inside Docker with ROS 2 Jazzy and the TurtleBot3 workspace sourced.

‘colcon build‘ result:

```
Summary: 3 packages finished

Packages:
  thermocator_msgs
  my_tb3_world
  thermocator
```

The full automated test run passed successfully.

‘colcon test‘ result:

100% tests passed, 0 tests failed out of 6

Tests passed:
  cppcheck
  flake8
  lint_cmake
  pep257
  uncrustify
  xmllint

HEADLESS DIGITAL TWIN SMOKE TEST

A headless launch of the digital twin Gazebo world was performed in Docker. Gazebo, ‘ros_gz_bridge‘, entity creation, and ‘robot_state_publisher‘ started successfully.

Observed Domain 1 topics during the smoke test:

/clock
/cmd_vel
/joint_states
/odom
/parameter_events
/robot_description
/rosout
/scan
/tf
/tf_static

Observed nodes:

/robot_state_publisher
/ros_gz_bridge

This confirms that the digital twin simulation can start and publish the basic Gazebo bridge topics required by the rest of the stack.

RUNTIME SYSTEM CHECKS

The user then started the full runtime setup through ‘scripts/bin/dock‘ and inspected topics from an attached Docker shell.

DOMAIN 38 ROBOT-SIDE TOPICS

Command:

ROS_DOMAIN_ID=38 ros2 topic list | grep -E ’^/(scan|odom|cmd_vel|thermocator/goals|thermocator/state/twinned)

Observed:

/cmd_vel
/odom
/scan
/thermocator/goals
/thermocator/state/twinned

This shows that the robot-side simulation/stack is active and that the bridged twin mission state and goal topic are present on Domain 38.

DOMAIN 1 DIGITAL TWIN TOPICS

Command:

ROS_DOMAIN_ID=1 ros2 topic list | grep -E ’^/(clock|scan|odom|cmd_vel|thermocator/goals|thermocator/state/loca

Observed:

/clock
/cmd_vel
/odom
/scan
/thermocator/state/local

The initial filtered check showed the core digital twin simulation topics and the local mission state bridged from Domain 38 into Domain 1.

DOMAIN 1 DIGITAL TWIN STACK NODES

Command:

```
ROS_DOMAIN_ID=1 ros2 node list | grep -E 'thermocator|decision|thermal|cartographer|planner|controller'
```

Observed:

```
/cartographer_node
/cartographer_occupancy_grid_node
/controller_server
/decision_node
/planner_server
/thermal_broadcaster
/thermal_map_builder
/thermocator_domain_bridge_1
```

This confirms that the digital twin is not only a Gazebo model. It is running its own SLAM, Nav2, thermal map, decision, and bridge-related components.

DOMAIN 1 MAPPING AND DECISION TOPICS

Observed relevant Domain 1 topics:

```
/action_map
/global_costmap/costmap
/goal_pose
/local_costmap/costmap
/map
/submap_list
/thermal_map
/thermal_reading
/thermocator/goals
```

This confirms that the digital twin stack produces map, thermal map, costmap, and goal candidate topics.

GOAL BRIDGE VERIFICATION

The most important digital twin communication check was whether goal candidates generated on Domain 1 appear on Domain 38.

DOMAIN 1 TWIN GOAL CANDIDATE

Command:

```
ROS_DOMAIN_ID=1 ros2 topic echo /thermocator/goals --once
```

Observed sample:

```
pose.header.frame_id: map
pose.pose.position.x: 1.3339702577560866
pose.pose.position.y: -0.2727715481081707
source: TWINNED
score: 93.53957250144425
```

This proves that the digital twin ‘decision_node‘ publishes ‘TWINNED‘ goal candidates on Domain 1.

DOMAIN 38 BRIDGED GOAL CANDIDATES

Command:

```
ROS_DOMAIN_ID=38 ros2 topic echo /thermocator/goals
```

Observed source sequence:

```
source: TWINNED
source: LOCAL
source: TWINNED
source: LOCAL
source: TWINNED
source: LOCAL
source: TWINNED
source: LOCAL
```

This is the key evidence that the digital twin goal stream is bridged into the robot-side domain. Domain 38 receives both local goal candidates and twin generated goal candidates on the same ‘/thermocator/goals‘ channel.

INTERPRETATION

The observed alternating ‘LOCAL‘ and ‘TWINNED‘ messages on Domain 38 mean:

* The Domain 38 ‘decision_node‘ publishes ‘LOCAL‘ goal candidates.
* The Domain 1 digital twin ‘decision_node‘ publishes ‘TWINNED‘ goal candidates.
* ‘domain_bridge‘ successfully forwards ‘TWINNED‘ goal candidates from Domain 1 to Domain 38.
* The robot-side ‘goal_arbiter‘ can compare local and twin-generated candidates.

CONCLUSION

The project implements a functional ROS 2 / Gazebo digital twin prototype. The digital twin runs an independent stack on Domain 1 and contributes decision outputs to the robot-side Domain 38 stack through ‘domain_bridge‘.

Verified status:

Build:	PASS
Automated tests/lint:	PASS
Digital twin launch:	PASS
Domain 1 stack active:	PASS
Goal bridge 1 -> 38:	PASS

Recommended report wording:

The system implements a ROS 2 based digital twin architecture in which a Gazebo-based twin runs an independent Thermocator stack and publishes TWINNED goal candidates. These candidates are bridged into the robot-side domain, where they appear alongside LOCAL candidates and can be evaluated by the goal arbiter.